

TRUONG TRONG HOA

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EDUCATION

Ha Long High School for the Gifted

Specialization: Informatics

Quang Ninh, Vietnam

Expected Graduation: May 2026

- **CGPA:** 8.5/10
- **Standardized Test:**
 - ◊ IELTS: 7.5 (Listening: 9.0, Reading: 8.0, Writing: 6.5, Speaking: 6.0)
 - ◊ SAT: 1400 (Verbal: 630, Math: 770)
- **Honors:**
 - ◊ Provincial Top Scorer in Quang Ninh – Specialized Informatics Entrance Exam
 - ◊ Merit Scholarship Recipient (2023–2024, 2024–2025)

ACADEMIC ACHIEVEMENTS

- **Gold Medal**, INNOVERSE Invention & Innovation Expo | 2025
- **3rd Prize**, Youth Informatics (Provincial) | 2025
- **Ranked 16th**, Quang Ninh Informatics National Team Selection | 2025
- **3rd Prize**, Vietnam Student Olympiad in Informatics (Provincial) | 2024
- **Consolation Prize**, Ha Noi Innovation Contest for Youth and Children | 2024

PUBLICATION

Artificial intelligence tools in the learning process of secondary school pupils. Vietnam Journal of Education, Jul 2025 (Issue 2), pp. 91–97. (**Co-author, Research Team Lead**)

- **Tech stack:** Python, TensorFlow/Keras, Google Colab, Google Drive, Matplotlib, PIL
- Led research scope on AI tools in education; coordinated dataset collection (~**1,000 samples**) and designed a case study on clean vs. contaminated vegetable classification.
- Implemented a CNN training and evaluation pipeline on Google Colab, including train/validation/test splits and reporting.

WORKING EXPERIENCE

Tan Phat Technology and Commerce Company Limited

Jun 2025 – Aug 2025

Software Development Intern

- **Tech stack:** Python, SQL, AWS (S3, Lambda, Glue), Hadoop, Git/GitHub
- Supported **Data Engineering** tasks by writing **Python** and **SQL** scripts for **data ingestion**, **cleaning**, and **reporting**.
- Built a personal **ETL pipeline** on **AWS** to **extract**, **transform**, and **load** datasets into structured outputs for downstream use.
- Worked with **distributed-data concepts** and tooling (**Hadoop ecosystem**) to handle larger datasets and **batch workflows**.

PROJECT EXPERIENCE

Software System for Supporting the Supervision of School Regulations and Discipline Oct 2025

Leader | [GitHub](#)

- **Tech stack:** Python, OpenCV, MediaPipe, YOLO, Google Colab, Vue.js
- Designed an end-to-end **monitoring workflow**: **data collection/labeling**, **model training**, and **deployment-ready inference**.
- Trained an **object detection** model and prepared **dataset configuration** for **reproducible training runs**.
- Implemented a lightweight **web UI** for **visualization** and system operation during demos.

AI-Driven Early Parkinson's Detection System

Aug 2025

Team Member / [GitHub](#)

- **Tech stack:** Python, MediaPipe, OpenCV, Jupyter/Colab
- Collected, curated, and **pre-processed** video samples to build a **training-ready dataset** (format **normalization**, **trimming**, **labeling**).
- Contributed to the **experimentation workflow** (feature extraction and **model iteration**) to support early detection signals.

Application of Artificial Intelligence in Detecting Food – Unsafe Vegetables

Jun 2025

Leader / [GitHub](#)

- **Tech stack:** Python, YOLO, TensorFlow, OpenCV, Google Colab, IoT sensors
- Developed an **object detection pipeline** to identify **unsafe vegetables** from images and video inputs.
- Integrated **IoT sensor readings** (physical/chemical measurements) to complement vision-based detection and improve **decision logic**.
- Set up cloud-based **training runs** and an iteration workflow for rapid experimentation.

SOCIAL ACTIVITIES

Organized charity for sick children at the National Institute of Hematology and Blood Transfusion

Organizer / Volunteer

- Led charity visits for pediatric patients; coordinated **volunteers** and **logistics** to deliver gifts and essential support.
- Donated blankets, school supplies, clothes, and toys; organized games and performances to improve **morale** during treatment.

“Flying Flower” Art Program

Volunteer

- Supported event operations for a city-wide **art & music festival**; helped coordinate community networking among clubs and visitors.

“Pink Heels – Warm Winter” Program at Dong Son Secondary School

Volunteer

- Co-led resource mobilization and distribution with total support value **VND 80,000,000**, benefiting disadvantaged students.
- Delivered **50 blankets + 50 gift sets + 1,000 books**; organized New Year cultural and sports exchanges; conducted home visits to encourage students to continue studying.

SKILLS

Programming Languages: C++, Python, C, Java, SQL

Machine Learning & AI: YOLOv5, OpenCV, MediaPipe, TensorFlow, PyTorch, Jupyter Notebook

Front-End: React, Vue.js, Node.js

Database & Cloud: Hadoop, AWS (S3, Lambda, Glue)

CERTIFICATION

- Python Basics – University of Michigan
- Python for Data Analysis: Pandas & NumPy – Coursera Project Network
- ChatGPT Advanced Data Analysis – Vanderbilt University
- Prompt Engineering for ChatGPT – Vanderbilt University